Supplementary Material for "Train on the Event, Deploy on the Baseline: Decoupling the Change Interval in Landslide Candidate Discovery"

S1-A. Implementation details

Encoders. Optical and SAR branches use Swin-B with out_indices (1, 2, 3) producing 256/512/1024 channels for optical and 192/384/768 for SAR. Topography (5 channels) and change (3 or 6 channels) use the lightweight modality stem: two stride-2 convolutions followed by three residual stages, producing the same three level widths. The fused representation is 256/512/1024 after per-level 1x1 projection.

Fusion. Per level, the projected modality features are stacked along the modality axis and passed through a 1x1 bottleneck (GELU) to per-modality logits, which are masked by the per-sample modality mask and softmaxed along the modality axis. The fused level is the gate-weighted sum. Gate logits are zero initialised.

Training. Eight epochs, AdamW, learning rate 1e-5, weight decay 1e-4, gradient clipping at norm 0.1, automatic mixed precision, batch size 4, resolution 320 (native patches are 128). Parameter-wise multipliers: backbone 0.5, modality encoders 1.0, fusion 1.0, head 1.0, router 1.0, meta-net 0.1, decoder 0.0, learnable text embedding 0.1. Only the most recent checkpoint is kept.

Verifier. MLP with hidden widths 256 and 64, listwise softmax objective over candidates of the same image, input 523 dimensions (frozen CLIP ViT-B/32 embedding plus 11 within-box statistics), Adam at 1e-3 with a step scheduler (step 10, gamma 0.7). Model-native scores are deliberately excluded from the input.

Interval intervention (Section V.D). Training stacks are built as delta NDVI, delta NDWI and delta NBR between the bracketing acquisitions, widened by one acquisition on each side. The deployment stack is unchanged from the long baseline. Only the training manifest's change paths change; the processor normalisation is recomputed from the training split.

S1-B. Full ablation (A0 to A9)

Table S1-B. Architecture ablation, AR@900. Reference values are from the 2026-10-12 batch; two configurations were re-run on 2026-10-20 with the published recipe (batch 4, eight epochs, seed 42) and are marked "re-run".

Arm	Modalities	Reference AR@900	Re-run AR@900
A0	optical	0.3533	**0.3714**
A1	optical + topography	0.4220	—
A2	optical + SAR	0.3745	—
A3	optical + SAR + topography	0.4574	—
A4	A3 + change	0.7167	0.7094 (anchor)
A5	A4 + modality tokens	0.6905	—
A6	A5 + cross-modal router	0.7059	—
A7	A6 without router	0.7029	—
A8	A6 without SAR	0.6729	—
A9	A6 without change	0.4799	**0.4337**

Reading A0. The re-run value lies within the run-to-run band of the reference (+0.018 against a band of 0.025), and its 95% CI, [0.350, 0.392], contains the reference value. A0 also reproduces the deployment-side picture: without change channels the strict-budget recall collapses to 0.092.

Reading A9. The re-run value is **below** the reference band (-0.046). The configuration was verified to apply the intended overrides (SAR and topography on, change off, modality tokens and router on) and to complete all eight epochs. A9 contains the two components that the paper reports as not improving performance, so its run-to-run spread is expected to be larger than A4's. We report the discrepancy rather than adjusting the arm, and treat the 2026-10-12 value as the archived reference. This is an open reproducibility gap for that single configuration.

S1-C. Candidate budgets and cost

Table S1-C1. Per-arm realised candidate budget, four-event split.

Arm	k=100	k=300	k=900	Mean realised budget
PF-RPN (export path)	0.3170	0.5908	0.6619	900
Fine-tuned YOLO-World	—	—	0.3321	736 to 842
Closed-set Y8	0.0617	—	0.3203	below 100 for one arm
Grounding DINO, zero-shot	—	—	0.1826	900
YOLO-World, zero-shot, fair	—	—	0.1019	736 to 842

Table S1-C2. False positives per image normalisation (10 FPPI). PF-RPN 0.1270; Grounding DINO 0.0019; fine-tuned YOLO-World 0.0021. Ratios 34.5x and 41.4x. Earlier drafts quoted 67x and 60x from a superseded normalisation; those values are retired and do not appear in the manuscript.

S1-D. Reproducibility of the criterion

Table S1-D. Per-fold criterion values and measured recall.

Fold	criterion (AUC-based)	measured AR@900	AR@900 gap
LOEO folds, 7	reported per fold in the release	—	—
LORO folds, 3	reported per fold in the release	—	—

Summary statistics: Pearson 0.915, Spearman 0.758, 95% CI [0.682, 0.986], leave-one-fold-out 0.856, sign agreement 10 of 10, largest single-fold influence 0.041, correlation with absolute recall -0.266 (CI crosses zero). Two-tier behaviour: AUC at or above 0.70 predicts a gap of 3.71 to 5.65 (17.7% to 26.9% of the detector's recall recovered), AUC at or below 0.50 predicts 1.37 to 1.57 (63.7% to 72.8%); no fold falls between 0.492 and 0.705.

S1-E. Change-channel interventions (all arms)

Table S1-E1. Same-tile permutation, anomaly-free controls (cloud, anchor 0.7059).

Arm	Operation	AR@900	Difference
clean	none	0.7059	anchor
change permuted	spatial permutation within tile	0.3866	-0.319
SAR permuted	spatial permutation within tile	0.6911	-0.015
topography permuted	spatial permutation within tile	0.6991	-0.007

Table S1-E2. Temporal interventions (cloud, anchor 0.7059).

Arm	Window	Span	AR@900	Difference
clean	event, first to last	1.00 T	0.7059	anchor

Arm	Window	Span	AR@900	Difference
temporal mispairing	two earliest acquisitions	about 0.02 T	0.3535	-0.352
equal-span control	outside window, maximum available	0.48 T	0.4632	-0.243

Strict equal-span control is not constructible on this dataset: the true pre/post span is the full series, and the annotated window lies in its middle. This is why the interval question is resolved by the calendar analysis of Section III.C rather than by span matching.

Table S1-E3. Inference-time substitution on a frozen model (RTX 4060, anchor 0.7085).

Change stack given to the model	Normalisation	AR@100	AR@900	Difference at 900
long baseline (matched)	model's own	0.4655	0.7085	anchor
event pair	model's own	0.2711	0.5460	-0.163
widened event window	model's own	0.3712	0.6217	-0.087
widened event window	variant statistics	0.3114	0.5941	-0.114
event pair	variant statistics	0.2362	0.5119	-0.197

Table S1-E4. Two intervals in one input (exploratory). A six-channel change stack that concatenates the long baseline and the widened event window was built for every tile, with the existing fusion gate left untouched, so that the three extra channels can be varied independently of the interval. Repeating the deployed long-baseline stack in those channels, and training and deploying on the long baseline, leaves recall on the anchor: AR@900 0.7041 against 0.7094 on the same machine (95% CI [0.6822, 0.7246]). Placing the event window in the same three channels, so that both intervals enter the model, reaches 0.7583 over two seeds (0.7634 and 0.7532, s.d. 0.0072), whose confidence interval does not overlap the control's. Against the widened-event training arm of the same seed, a paired per-image bootstrap gives +0.049 at k=100 (95% CI [+0.025, +0.073]) and +0.029 at k=300 ([+0.008, +0.050]), but +0.007 at k=900 ([-0.011, +0.026]). Unlike the widened-event arm, this configuration needs the event stream at inference, so it is reported as an exploratory extension rather than as part of the free intervention (Table S1-E5).

Table S1-E5. Six-channel arms (exploratory). The control repeats the deployed long-baseline stack in the three extra change channels; the two-interval arm places the widened event window there. All arms are scored with 900 candidates per image on the four-event test split (1,759 images) on the RTX 3080 Ti, whose own anchor is 0.7094; the archived anchor of 0.7059 sits 0.005 lower, inside the run-to-run noise of 0.0059.

Arm	AR@100	AR@300	AR@900
six channels, both intervals long (control)	0.4384	0.5857	0.7041
six channels, long + event window, seed 42	0.5789	0.6867	0.7634
six channels, long + event window, seed 123	0.5607	0.6783	0.7532
mean and sd (two seeds)	0.5698 ± 0.0129	0.6825 ± 0.0059	0.7583 ± 0.0072

S1-F. Interval analysis (pixel level and calendar)

Table S1-F1. Pixel-level separability by window. Four test events, about 400 annotated tiles, official masks as ground truth, paired bootstrap over tiles.

Window	Span	AUC	recall@1%FPR	usable pixels	delta AUC vs long baseline
long baseline	360 d	0.7771	0.1948	0.982	—
event pair	10 d	0.7439	0.1189	0.776	-0.045 [-0.071, -0.055]
event, widened by one	25 d	**0.8476**	**0.2704**	**0.985**	**+0.066 [+0.046, +0.060]**
event, widened by two	70 d	0.8315	0.2742	0.987	+0.051 [+0.031, +0.044]
widened, validity-gated	70 d	0.8303	0.2709	0.990	+0.050 [+0.028, +0.044]
same span, displaced	25 d	0.5064	0.0155	0.960	-0.240 [-0.273, -0.256]
label-free NDVI-drop rule	10 d	0.5147	0.0300	0.875	-0.259 [-0.290, -0.270]

Table S1-F2. Calendar reproduction of the annotated window, per inventory.

Inventory	n	exact	within one acquisition
italy	1,799	1.000	1.000
chimanimani	345	0.986	1.000
hiroshima	324	1.000	1.000
kyrgyzstan1	186	0.823	1.000
hokkaido	144	1.000	1.000
newzealand	94	0.926	1.000
itogon	82	1.000	1.000
kyrgyzstan2	58	0.759	1.000
indonesia	47	0.979	1.000
china	28	0.821	1.000
usa	28	1.000	1.000
all	**3,135**	**0.979**	**1.000**

S1-F4. Window properties and the gain, by protocol

Table S1-F4. Window properties and the interval gain, by protocol. Medians over the annotated tiles of each split; "tiles with dates" counts the tiles whose annotation carries a pre/post pair, and only those contribute to the window columns. The last two rows score the same 253 Indonesian tiles and differ only in the backing model, so the sign of the gain is set by the protocol rather than by the region.

Split / event	GT tiles	cadence (d)	event ±1 span (d)	usable long / pair / ±1	anchor AR@900	±1 AR@900	Δ AR@900
four-event test, chimanimani	345	22	25	0.999 / 0.851 / 1.000	0.7091	0.7706	+0.0615
four-event test, china	47	101	175	0.926 / 0.000 / 0.979	0.7963	0.8102	+0.0138
four-event test, indonesia	253	88	170	1.000 / 0.179 / 1.000	0.7137	0.7551	+0.0414
four-event test, usa	28	28	50	0.991 / 0.822 / 0.995	0.5284	0.4991	-0.0294
leave-one-region-out fold, indonesia	253	88	170	1.000 / 0.179 / 1.000	0.7891	0.7717	-0.0173

The last two rows of Table S1-F4 hold the region, its acquisition calendar and its windows fixed and change only the backing model: under the event-level split the same 253 Indonesian tiles gain 0.0414 at AR@900, under the leave-one-region-out fold they lose 0.0173. Window validity is not the difference, because the widened window is fully usable in both (1.000), and neither is cadence (88 d in both, an event window of 170 d). What differs is how strong the baseline already is on that region, 0.7137 against 0.7891, which is consistent with the gain being concentrated where candidates are scarce. The intervention therefore does not replicate under the second protocol, and Section VI-E of the paper reports it as an open limitation rather than as a boundary that has been explained away. Note also that the annotated pair is almost entirely unusable in Indonesia and china (0.179 and 0.000), which is the same effect that makes the single adjacent pair the weakest window in Table S1-F1.

Table S1-F3. Distance from the event date to the bracketing acquisitions. Previous: median 3 days, 90th percentile 5 days. Next: median 4 days, 90th percentile 7 days. The annotated window is a single adjacent pair in 100% of tiles (median span one acquisition), positioned with a median relative index of 0.50.

S1-G. Annotation-date quality

Table S1-G. Multi-event tiles among the 4,392 positive tiles.

Candidate dates	n	share	median GT pixels	median date confidence
single window	3,139	71.5%	224	1.000
2	366	8.3%	245	0.444
3	273	6.2%	385	0.464
4	177	4.0%	446	0.451
5	109	2.5%	651	0.498
6	83	1.9%	753	0.488
7	96	2.2%	1,110	0.526
8	61	1.4%	1,255	0.579
9	33	0.8%	1,907	0.575
10	27	0.6%	2,018	0.599
11 or more	28	0.6%	2,510 (at 11)	0.610

Rule used for multi-event tiles: the highest-confidence candidate (first entry of the ranked list). Unannotated tiles: the neighbouring event's calendar, or the series midpoint when no event date exists.

S1-H. Reproduction and validation

Table S1-H1. Data and pipeline checks.

Check	Result
Tile and array count	10,015 tiles, 60,090 arrays, 0 inconsistencies
Long-baseline reproduction	40 tiles across 4 events, maximum absolute difference 0, masks identical 40 of 40
Anchors by machine	0.7059 (cloud), 0.7085 (RTX 4060), 0.7094 (RTX 3080 Ti)
Run-to-run envelope per fold	0.0076, 0.0234, 0.0251
QA gate	0 flagged artefacts

Table S1-H2. Claims added in this revision and their artefacts.

Claim	Value	Artefact
Calendar reproduction	0.979 / 1.000, n=3,135	calendar rule output
Pixel-level window AUC	0.7771 / 0.7439 / 0.8476 / 0.5064	window mechanics table
Frozen substitution	0.7085 to 0.6217 / 0.5460	substitution evaluations
Training-axis ordering	0.6969 / 0.7094 / 0.7563	Table Z
Paired gains at three budgets	+0.0644 / +0.0694 / +0.0468	per-image dumps, paired bootstrap

Claim	Value	Artefact
Multi-event share	0.285	feature dump
Channel-count control	0.7041 against 0.7094 on the same machine	6-channel duplicated-stack run
Two-interval representation	0.7583 ± 0.0072, two seeds	6-channel arm, seeds 42 and 123
Cross-protocol fold (negative)	0.7891 to 0.7717 at AR@900; -0.0423 at AR@300	leave-one-region-out run, paired bootstrap

S1-I. Re-run values for the ablation table

Table S1-I. Re-runs performed for this revision. All carry batch size 4, eight epochs, seed 42, and the published recipe.

Configuration	Reference	Re-run	Difference	Status
A0	0.3533	0.3714	+0.018	within band, CI contains reference
A9	0.4799	0.4337	-0.046	**outside band, open discrepancy**
A4 (anchor)	0.7059	0.7094	+0.0035	within band

Seed replicates of the widened-event training arm. Seeds 42, 123 and 777 were run on the identical data definition and are evaluated on the deployed long-baseline root, so only the training interval differs from the anchor. The earlier seed-42 checkpoint was trained before the multi-event window rule was finalised, which is why seed 42 is repeated here.

Seed	AR@100	AR@300	AR@900
42	0.5165	0.6456	0.7594
123	0.5127	0.6459	0.7400
777	0.5299	0.6551	0.7429
mean and sd	0.5197 ± 0.0091	0.6488 ± 0.0054	0.7474 ± 0.0105

The three seeds give AR@900 0.7594, 0.7400 and 0.7429, a mean of 0.7474 with a seed standard deviation of 0.0105; at k=300 the mean is 0.6488 (s.d. 0.0054) and at k=100 0.5197 (s.d. 0.0091). Combining the target's seed variance with the anchor's run-to-run variance of 0.0059 puts the AR@900 gain at 5.5 standard errors, the AR@300 gain at 10.6 and the AR@100 gain at 8.6; a paired per-image bootstrap on the first seed gives +0.0468 (95% CI [+0.029, +0.065]) at k=900.

S1-J. Pre-registered test on the weakest fold

Why this test. The leave-one-region-out fold on Indonesia loses 0.0173 AR@900 under the widened-event window, whereas the event-level split gains 0.0414 on the same 253 Indonesian tiles; the two protocols hold the region, its calendar and its windows fixed and differ in the backing model, whose anchor is 0.7137 against 0.7891 on that subset (Table S1-F4). One reading is that the intervention pays in proportion to the headroom a model leaves on the deployment region. This section fixes the test of that reading, and its decision rule, before the runs finish.

Prior evidence that does not favour the account. The same four-event split gives no such gradient: across its four events the anchor level does not predict the gain (Pearson +0.65 on the values and +0.20 on ranks), the event with the lowest anchor (usa, 0.528) is the one that loses most (-0.0294 at AR@900), and the event with the highest anchor (china, 0.796) gains only +0.0138. The account is therefore tested as a candidate explanation that the existing evidence does not support, which is why a positive outcome would be reported as surprising rather than as confirmation.

Design. Italy is the fold with the weakest archived anchor (AR@900 0.2896, re-scored here at 0.2950, a difference inside the run-to-run envelope). The widened-event arm is trained for eight epochs on the fold's identical 5,716-tile training set with seeds 42, 123 and 777, and evaluated on the fold's long-baseline root; the archived anchor weights are re-scored on the same root. Preprocessing, the recomputation of the change-channel normalisation from the training split, and the evaluation protocol follow Section V-D, so the only difference from the anchor is the change stack used in training. No leave-one-out fold has an anchor between 0.5 and 0.6 - the archived anchors are bimodal, 0.2896 and then 0.6757 to 0.7889 - so a multi-point dose-response gradient cannot be assembled from the archived folds and a single fold is what this design can supply.

Decision rule, fixed in advance. Let dAR@k be the seed-averaged per-image AR@k minus the anchor's per-image AR@k, and let the decision interval be the 95% percentile interval of a paired bootstrap over the test images that carry ground truth (20,000 resamples). At k = 900: (a) if the upper bound is at or below 0, the headroom account is dropped from the paper and the result is reported here as a negative control; (b) if the lower bound is above 0, the account is reported as consistent with the data and not as confirmed, because one fold cannot separate headroom from event difficulty, annotation quality or domain shift; (c) if the interval contains 0, the account is downgraded to a speculation and the paper states only the conditional form.

Reporting obligation. Per-seed intervals are reported alongside the seed-averaged interval, and the seed dispersion is combined with the anchor's run-to-run standard deviation of 0.0059 in the same way as Section V-D. Whatever the outcome, the test is reported: a null or negative result enters this document and one sentence of Section VI-E rather than being withheld. What the rule freezes is the claim, not the data.

S1-K. Result of the pre-registered test (negative control)

Outcome. The widened-event arm loses recall on the weakest fold at all three budgets, in all three seeds. The seed-averaged per-image AR@900 falls from 0.2950 for the anchor to 0.2280, a difference of -0.0670 with a paired-bootstrap 95% interval of [-0.0734, -0.0607]; AR@300 falls from 0.2004 to 0.1484 (-0.0520, [-0.0576, -0.0465]) and AR@100 from 0.1392 to 0.0944 (-0.0448, [-0.0498, -0.0400]). Per seed at AR@900: 42 -0.0648 [-0.0721, -0.0575], 123 -0.0584 [-0.0656, -0.0511], 777 -0.0779 [-0.0859, -0.0700]; the seed-to-seed standard deviation of the difference is 0.0099, against an anchor run-to-run standard deviation of 0.0059. Every interval lies below zero, so clause (a) of the decision rule applies: the headroom account is dropped from the paper, and the result is recorded here as a negative control.

Why this fold fails, as far as the data reach. The loss is not a matter of window geometry or of validity on the widened side. The tiles of this fold have a deployed baseline of 340 d in the median against 365 d on the four-event split, the same 25 d cadence, a widened window of 30 d against 25 d, and 0.977 of their pixels usable in that window. Two things are unusual there: the annotated pair itself is unusable in 0.000 of its pixels, and the fold is the one dominated by the smallest landslides in the benchmark, since its recall at 900 proposals is the lowest of the ten folds (AR@900 0.2896 archived). A window widened to 30 d would therefore mix a bare-soil signal with the re-growth of objects a few pixels wide. That reading is consistent with the size of the loss, but this test does not establish it, and we report it as the most plausible of several explanations rather than as a mechanism.

Consequence for the paper. The paper keeps the account out of its claims, states the gain as a result of the event-level protocol with its own anchor, and records in Section VI-E that a fold chosen for maximal headroom did not benefit. The three specification claims are untouched: the interval is a calendar fact (S1-F), it cannot be corrected after training (S1-E3), and committing it during training leaves the deployed input

unchanged (Section V-D). This test manipulated only the training window, so a failure here bounds the benefit of the intervention without bearing on those three statements.